



Reduced Boundary Sensitivity and Improved Contrast of the Regularized Inverse Problem Solution in Elasticity

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We observe that posing the inverse problem as a constrained minimization problem under regularization leads to boundary dependent solutions. In this paper, we propose a modified objective function and show with 2D examples that our method works well to reduce boundary sensitive solutions. The examples consist of two stiff inclusions embedded in a softer unit square. These inclusions could be representative of tumors, which are in general stiffer than their background tissues, thus could potentially be detected based on their stiffness contrast. We modify the objective function for the displacement correlation term by weighting it with a function that depends on the strain field.

In a simplified 1D coupled model, we derive an analytical expression and observe the same trends in the reconstructions as for the 2D model. The analysis in this paper is confined to inclusions of similar size and may not overlap when projected on the horizontal axis. They may, however, vary in position along the vertical axis. Furthermore, our analysis holds for an arbitrary number of inclusions having distinct stiffness values. Finally, to increase the overall contrast of the tumors and simultaneously improve the smoothness, we solve the regularized inverse problem in a posterior step, utilizing a spatially varying regularization factor. [DOI: 10.1115/1.4031937]

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1 Introduction

Modeling observational data from known model parameters has been well established for problems in elasticity [1–3]. For example, knowing the material property distribution for a linear elastic solid, such as the Young’s modulus and Poisson’s ratio, we can predict its deformation field under various prescribed boundary conditions. This is often referred to as the forward problem and can be solved numerically using finite element, finite difference, or finite volume methods. On the other hand, observational data can be measured and utilized to determine the spatial distribution of the model parameters, which is referred to as an inverse problem. For problems in elasticity, the deformation of the body can be observed utilizing medical imaging tools, such as ultrasound devices [4–15], magnetic resonance imaging (MRI) [16–23], and optical coherence tomography (OCT) [24,25].

Solving the inverse problem in elasticity for the elastic parameter distribution has important applications in diagnosing diseased tissues based on their elastic parameter distribution, as diseased tissues change their elastic parameters from normal due to changes in their microstructure [26]. For example, the feasibility to distinguish benign and malignant breast tumors has initially been shown utilizing strain imaging [8,10,13,27–30], where the axial strain is interpreted to be inverse proportional to the Young’s modulus. This method has been extended to imaging atherosclerotic plaques [31–33]. The advantage of strain imaging lies in its simple inversion, and the elastic

parameter distribution can be determined in quasi-real time using ultrasound techniques. However, in strain imaging it is assumed that the stress is constant in the problem domain, which is not true for a nonhomogeneous solid with nonuniform loading. Thus, the representation of tissue elasticity through strain images could result in undesired artifacts.

Despite this approximation, strain imaging has shown potential in detecting tumors based on their stiffness contrast. This motivated researchers to solve for the actual elastic parameter distribution directly from the equations of equilibrium [34,35]. This approach requires gradients of noisy displacement field, thus amplifying the noise level drastically in the strain field [36], which is utilized in the constitutive relationship of the material. To circumvent taking derivatives of noisy data and use incomplete interior displacement data, the inverse problem has been posed as a constrained minimization problem, where the difference between a measured and computed displacement field is minimized in the L-2 norm under the constraint that the computed displacement field satisfies the forward problem [37,38]. This formulation has been further improved by augmenting the minimization function with a total variation regularization term [39–45]. The constrained minimization problem has been solved successfully using a gradient-based minimization method, in particular the limited BFGS method [46–48], which requires the gradient of the objective function with respect to the unknown elastic parameters at each minimization step. The gradient has been determined efficiently using the adjoint equations [49], reducing the computational time of the inverse solution significantly. Nevertheless, this method is computationally extensive and the inverse problem cannot be solved in quasi-real time, which is of high importance in a clinical environment. The regularized inverse problem has been successfully tested on noisy hypothetical data [6,39,41,43,50], tissue mimicking phantoms [6,39], and breast tumor patients [40,42].

Recently, we realized that this inverse strategy to solve for the shear modulus distribution depends on the deformation field. In other words, the shear modulus reconstruction is sensitive to changing boundary conditions for a given problem domain. In the present paper, we demonstrate that this issue occurs when two

inclusions, e.g., representing two tumors, in a homogeneous background are present. We illustrate that the strain field and the total variation (TV) regularization play a key role in the elastic parameter reconstruction. We address this issue utilizing a spatially weighted displacement correlation term as a function of the strain field. We thoroughly discuss these observations in Sec. 4 and derive a coupled 1D analytical expression to analyze our findings. We observe that boundary sensitivity (1) occurs independent of noise and (2) can be reduced by modifying the displacement correlation term by a strain dependent term. The 1D results closely resemble the results obtained from the 2D continuum model. Our analysis shows that the modified objective function works for any number of similar sized inclusions embedded in a soft background with distinct inclusion stiffness values. This analysis however is constrained by the fact that the inclusions may not share the same horizontal position, while the vertical position of each inclusion may be arbitrary. Furthermore, we improve the contrast of the elastic parameter reconstructions utilizing a spatially varying regularization factor in a posterior step, while at the same time yielding a smoother elastic parameter reconstruction.

2 Methods

Throughout this paper, we confine our view to a linear elastic material in two-dimensional space. For most human tissues, this is only true if the deformations are small, e.g., about 1% strain. Furthermore, we model the solid in plane strain and assume material incompressibility, as most soft human tissues mainly consist of fluids, thus, are nearly incompressible. In the following, we define three problem domains with unit square length and distinct target shear modulus distributions to create hypothetical “measured” displacement data by solving the forward problem for static deformations. More precisely, we use mixed finite element methods that can be derived from the strong form given by: Find the displacement field $\mathbf{u}$ and the pressure p such that

$$\text{div}(\boldsymbol{\sigma}) = 0 \quad \text{in } \Omega \quad (2.1)$$

$$\mathbf{u} = \mathbf{g} \quad \text{on } \Gamma_g \quad (2.2)$$

$$\boldsymbol{\sigma} \cdot \mathbf{n} = \mathbf{h} \quad \text{on } \Gamma_h \quad (2.3)$$

The incompressibility constraint is enforced by

$$\text{tr}(\boldsymbol{\varepsilon}) = 0 \quad \text{in } \Omega \quad (2.4)$$

In Eq. (2.1), Ω denotes the problem domain and $\boldsymbol{\sigma}$ is the Cauchy stress tensor. In Eqs. (2.2) and (2.3), Γ_g and Γ_h denote the Dirichlet and Neumann boundary conditions, respectively, where $\mathbf{g}$ defines the prescribed displacement boundary and $\mathbf{h}$ defines the traction boundary with unit outward normal $\mathbf{n}$. Furthermore, these boundaries must satisfy the conditions that $\Gamma_h \cup \Gamma_g$ represents the closed boundary of the problem domain and $\Gamma_h \cap \Gamma_g = \emptyset$. Finally, the incompressibility condition in Eq. (2.4) requires that the trace of the strain tensor $\boldsymbol{\varepsilon}$ is zero.

To solve the forward problem, we used a uniform mesh with hybrid finite elements and interpolated the displacement and the pressure variables with linear triangular elements. We used stabilized finite elements to circumvent the Ladyzhenskaya-Babuska-Brezzi conditions according to Refs. [51] and [52]. Finally, we add about 3% white Gaussian noise to the displacement field to be more realistic with measured noisy displacement data when solving the inverse problem. We emphasize that for the mesh utilized in this paper, this corresponds to about 75% strain noise for the given mesh size.

We reconstruct the target shear modulus distribution utilizing the conventional method to solve the regularized inverse problem (Sec. 2.1) and show that it is sensitive to changing boundary conditions (Sec. 3.1). “Conventional” method here refers to the regularized inverse problem formulation posed as a constrained minimization problem utilized in Refs. [6,39,41,42,53].

We introduce the novel inverse method in Secs. 2.2 and 2.3, and show in Secs. 3.2 and 3.3 that the shear modulus reconstructions improve significantly. The problem domains are given as follows: In Fig. 1(a), we have two inclusions positioned horizontally in a homogeneous background, with a shear modulus of 5 in both inclusions and 1 in the background. The bottom edge is fixed in vertical direction and the center node on that edge is fixed in the horizontal direction to avoid rigid body motion. On the top edge, we apply a vertical displacement boundary, in case 1 with a uniform displacement distribution and in case 2 with a linear vertical displacement boundary. The reason behind this is to create the same strain state in both inclusions for the uniform displacement boundary and a distinct strain state in both inclusions for the linear displacement boundary. We apply a displacement of 0.01 units for the uniform displacement boundary, and $u = 0.002 + 0.008x$ for the linear displacement boundary, where $x = 0$ is located at the left vertical edge. All other nondefined boundaries are traction free. In Fig. 1(b), we have a similar problem domain to Fig. 1(a), and the only difference is that the shear modulus values in the inclusions are distinct. The left inclusion has a shear modulus value of 5 and the right inclusion has a shear modulus value of 10. A uniform displacement boundary is applied on the top edge, and all other remaining boundary conditions are the same as given for the problem domain in Fig. 1(a). We note that the distinct stiffness values in the left and right inclusion will yield distinct strain states. Despite the uniform compression, we will observe in later sections that the shear modulus reconstruction for this example is ambiguous. Finally, in Fig. 1(c) two inclusions are positioned on the diagonal of the unit square, where the shear modulus in both inclusions is 5 and the shear modulus in the background is 1. The boundary conditions are the same as for the problem domain given in Fig. 1(a), case 2 with linear displacement boundary. The boundary conditions utilized in solving the inverse problem are the same as those in the corresponding forward

[Figure 1 — placeholder]

Three-panel target shear modulus distribution color maps. (a) Two horizontally positioned circular inclusions with $\mu = 5$ in a homogeneous background of $\mu = 1$. Downward arrows “u” on top boundary indicate uniform or linear displacement loading; bottom boundary is fixed (hatched). Colorbar range SM = 1–5. (b) Two horizontally positioned inclusions with $\mu = 5$ (left) and $\mu = 10$ (right) in background $\mu = 1$. Colorbar range SM = 1–10. (c) Two inclusions positioned on the diagonal of the unit square, both $\mu = 5$, background $\mu = 1$. Linear displacement boundary on top.

Fig. 1 Given are the following target shear modulus distributions: (a) two horizontally positioned inclusions with $\mu = 5$ in a homogeneous background of $\mu = 1$, (b) two horizontally positioned inclusions with $\mu = 5$ in the left inclusion, $\mu = 10$ in the right inclusion, and $\mu = 1$ in the background, (c) two inclusions positioned on the diagonal of the unit square, with $\mu = 5$ in both inclusions and $\mu = 1$ in the background

problem except that noise is introduced in the displacement boundary conditions in the inverse problem.

We note that we do not prescribe any nonzero traction boundary conditions, thus the shear modulus reconstructions are determined only up to a multiplicative factor. This poses no difficulties in solving the shear modulus distribution utilizing total variation regularization as being discussed in [40,50].

2. of the Iterative, Regularized Inverse Problem Formulation.

we begin with a brief review of the conventional regularized inverse problem statement [40,41,53]: Given the small measured displacement fields $\mathbf{u}_{\text{meas}}^1, \mathbf{u}_{\text{meas}}^2, \dots, \mathbf{u}_{\text{meas}}^n$, find the shear modulus distribution μ , such that the objective function

$$F = \frac{1}{2} \sum_{i=1}^n \|\mathbf{T}(\mathbf{u}^i - \mathbf{u}_{\text{meas}}^i)\|_2^2 + \frac{1}{2} \alpha \int_{\Omega} \sqrt{|\nabla \mu|^2 + c^2} d\Omega \quad (2.5)$$

is minimized subject to the constraint that the computed displacement fields $\mathbf{u}^i$ satisfy the forward elasticity problem. The upper index i denotes the i th displacement field in Eq. (2.5) and the summation indicates that we can accommodate multiple displacement fields into the inverse problem formulation. The first term is the displacement correlation term, minimizing the difference between the computed and measured displacement fields in the L-2 norm. The tensor $\mathbf{T}$ is a diagonal tensor that assigns different weights to each displacement component. This is important utilizing displacement data obtained from ultrasound techniques, because estimates of the lateral displacement component (perpendicular to the transducer axis) are low precision relative to the axial displacement component (parallel to the transducer axis). Thus, one may choose the diagonal tensor such that highly noisy displacement components contribute less (or even not at all) to the overall objective function. In this paper, we will omit the noisy displacement component by setting the first diagonal component of the $\mathbf{T}$ tensor to zero and the second diagonal component to one. The second term in Eq. (2.5) is the regularization term that can be thought of as a penalty term to penalize oscillations and smooth the final solution spatially. In particular, the regularization term being utilized is the total variation, which has the advantage that it preserves sharp interfaces of objects, e.g., interface between tumor and background. The weight of the regularization term is controlled by the regularization factor α , which is noise dependent, i.e., the greater the noise level the higher the regularization factor. It can be determined using the L-curve method [54–57], based on Morozov’s discrepancy principle [58–60], or based on smoothness criteria. In this paper, we will assume that a small region in the problem domain is smooth, which is a reasonable assumption for most soft tissues. Thus the regularization factor will be chosen such that some small region in the problem domain is sufficiently smooth, but not overly smoothed. To quantify smoothness, one could determine the standard deviation in some small and smooth region (with no object interfaces) comprising a submesh of 5×5 nodes, assuming that the mesh resolves the shear modulus distribution well. The optimal lower and upper threshold for the standard deviation is problem type dependent and will guide in the selection of the optimum regularization factor. In all our reconstructions, we have started with a small regularization factor and increased it incrementally until the desired smoothness has been reached. Finally, c denotes a small constant to avoid singularities in the gradient computation.

The constraint optimization problem is solved using the limited BFGS method, and the gradient is evaluated efficiently using the adjoint method. This requires to solve one forward problem and

one adjoint problem at each minimization call. The forward problem is solved utilizing finite element techniques, and the unknown shear modulus is interpolated with the same finite element shape functions utilized for the displacement and pressure variables. Details on solving the inverse problem using the adjoint method have been thoroughly discussed in Refs. [41] and [61] and will not be reviewed here.

2.2 Spatia Weighted Displacement Correlation Method.

The proposed inverse formulation involves a modification of the objective function given as

$$F = \frac{1}{2} \sum_i^n \|\mathbf{W}^i(\mathbf{u}^i - \mathbf{u}_{\text{meas}}^i)\|_2^2 + \frac{1}{2} \alpha \int_{\Omega} \sqrt{|\nabla \mu|^2 + c^2} d\Omega \quad (2.6)$$

where $\mathbf{W}^i$ is a weighting function of the measured normal strain components ε_{xx}^i , ε_{yy}^i , and ε_{zz}^i corresponding to the i th deformation field

$$\mathbf{W}^i = \begin{pmatrix} \frac{1}{\varepsilon_{xx}^i} & 0 & 0 \\ 0 & \frac{1}{\varepsilon_{yy}^i} & 0 \\ 0 & 0 & \frac{1}{\varepsilon_{zz}^i} \end{pmatrix} \quad (2.7)$$

As discussed earlier, we assume that the deformation field is in a two-dimensional space, thus we omit the displacement and strain components in the z -direction. Furthermore, assuming that we have only one “measured” displacement field, and that the displacement component in the x direction is highly noisy and does not contribute to the objective function (see Sec. 2.1), Eq. (2.6) reduces to

$$F = \frac{1}{2} \left\| \frac{(\mathbf{u}_y - (\mathbf{u}_{\text{meas}})_y)}{\varepsilon_{yy}} \right\|_2^2 + \frac{1}{2} \alpha \int_{\Omega} \sqrt{|\nabla \mu|^2 + c^2} d\Omega \quad (2.8)$$

We note that we do not utilize the measured strain field directly in Eqs. (2.6) and (2.8). We spatially weight the displacement correlation term, as it contains highly amplified noise levels. We have decided to utilize the strain field from the final computed displacement field after solving the inverse problem according to the method presented in Sec. 2.1. An alternative approach would be to filter the noisy “measured” strain field, but from our experience the computed strain field yields superior quality in the reconstructed shear modulus distribution. We will investigate the modified objective function further in Sec. 4 and compare this to the conventional inverse formulation.

2.3 Spatially Varying Regularization Factor.

We solve the inverse problem in a posterior step utilizing the shear modulus distribution from the solution of the inverse problem with the spatially weighted displacement correlation term according to Sec. 2.2. To this end, we express the regularization factor in terms of the shear modulus distribution obtained utilizing Eq. (2.8). The new regularization factor is given by

$$\bar{\alpha} = \frac{\alpha}{\mu} \quad (2.9)$$

Thus, the new regularization factor is a function of the spatial coordinates

$$F = \frac{1}{2} \left\| \frac{(\mathbf{u}_y - (\mathbf{u}_{\text{meas}})_y)}{\varepsilon_{yy}} \right\|_2^2 + \frac{1}{2} \int_{\Omega} \bar{\alpha} \sqrt{|\nabla \bar{\mu}|^2 + c^2} d\Omega \quad (2.10)$$

We denote the unknown shear modulus distribution in Eq. (2.10) by $\bar{\mu}$ to distinguish it from the known shear modulus in Eq. (2.9). In Ref. [62], a spatially treated regularization term has been presented for atherosclerotic plaques with known nonhomogeneous morphology of the plaque in the artery from spatial priors.

3 Results

The limited BFGS method is utilized to solve the constrained optimization problem, which requires the computation of the gradient of the objective function with respect to the unknown elastic parameter distribution at each minimization call. We will stop the minimization procedure when fully converged, thus the total number of iterations is about 3000–6000 in the proceeding results. All reconstructions in Sec. 3 utilize the displacement field, determined from the forward problem for the boundary conditions given in Fig. 1 after adding about 3% of white Gaussian noise. The regularization constant c for all these reconstructions is chosen to be 0.01. The initial guess of the shear modulus is 1 over the whole domain, and the lower and upper bounds for the search region of the unknown shear modulus distribution are between 1 and 30, respectively.

3.1 Results for the Iterative, Regularized Inverse Problem.

In this section, we present results for the regularized inverse problem using the methodology reviewed in Sec. 2.1 and the examples from Sec. 2. Figure 2 represents the shear modulus reconstructions for the target shear modulus distribution given in Fig. 1(a) for the case with a uniform compression. In Fig. 2, the left reconstruction corresponds to a regularization factor of 8.0×10^{-10} and the right reconstruction corresponds to a regularization factor of 2.5×10^{-10} . Clearly, both inclusions are well recovered as well as their shear modulus values. The plots on the bottom of Fig. 2 represent the shear modulus values along a horizontal line through the center of both inclusions for each shear modulus reconstruction. The horizontal centerline plot for the target shear modulus distribution is included in these plots for comparison.

Now, we demonstrate that the conventional solution of the regularized inverse problem, posed as a constrained minimization problem, leads to undesired shear modulus reconstructions when boundary conditions are altered to a linear displacement compression. The shear modulus reconstruction is given in Fig. 3 for a regularization factor of 5.0×10^{-10} (left column) and 1.2×10^{-10} (right column). We observe that the shape of the inclusions are well recovered, however, the shear modulus values within each inclusion differ significantly from each other. Clearly, the current solution of the inverse problem appears to be dependent on the choice of boundary conditions or similarly related to the displacement field. We will elaborate on this further in Sec. 4 and show for a set of shear modulus distributions that the solution of the inverse problem is in fact sensitive to the strain distribution.

[Figures 2, 3, 4 — placeholders]

Fig. 2: Two-panel shear modulus reconstruction for Fig. 1(a) with *uniform* compression, two regularization factors. Top: color maps. Bottom: horizontal centerline plots (reconstruction vs. target). Both inclusions well recovered.

Fig. 3: Two-panel shear modulus reconstruction for Fig. 1(a) with *linear* compression, two regularization factors. Top: color maps showing unequal inclusion values. Bottom: centerline plots showing the asymmetry.

Fig. 4: Two-panel shear modulus reconstruction for Fig. 1(b) with *uniform* compression, two regularization factors. Target: left $\mu = 5$, right $\mu = 10$. Right inclusion significantly underestimated despite uniform loading.

In Fig. 4, we reconstruct the target shear modulus distribution given in Fig. 1(b) for a displacement field determined under a uniform compression. The location of the inclusions is the same as before, but the right inclusion is by factor of 2 stiffer than the left inclusion. The reconstructed shear modulus distribution in Fig. 4 is given for a regularization factor of 1.0×10^{-10} (left plot) and a regularization factor of 5.0×10^{-11} (right plot). We observe that the shape of the inclusions is well recovered, while the reconstructed shear modulus values between the left and right inclusion differ significantly with respect to their contrast loss. For example, comparing the horizontal centerline plots between the target and reconstructed shear modulus distribution, we observe that the shear modulus value in the left inclusion is well recovered, while the shear modulus value in the right inclusion is significantly underestimated (Fig. 4, left column). In other words, the relative loss in contrast of left inclusion to background versus right inclusion to background is different, despite the fact that a uniform compression is utilized. This leads to a discussion on the effect of strain rather than the displacement field causing these undesired artifacts (see Sec. 4).

In Fig. 5, we demonstrate that the conventional solution of the regularized inverse problem, posed as a constrained minimization problem, also fails to yield proper reconstructions of the shear

[Figures 5, 6, 7, 8, 9 — placeholders]

Fig. 5: Shear modulus reconstructions for Fig. 1(c) with *linear* compression, two regularization factors (4.0×10^{-10} and 1.0×10^{-10}). Inclusions on diagonal. Left inclusion clearly underestimated. Centerline plots along diagonal.

Fig. 6: Shear modulus reconstruction for Fig. 1(a) using the spatially weighted method (Sec. 2.2) with linear compression. Regularization factor 3×10^{-5} . Difference between inclusions significantly reduced. Centerline plot comparison.

Fig. 7: Reconstruction for Fig. 1(b) with uniform compression using the spatially weighted method. Regularization factor 7.0×10^{-6} . Inclusions well recovered, relative contrast loss more consistent.

Fig. 8: Reconstruction for Fig. 1(c) with linear compression using the spatially weighted method. Regularization factor 2.5×10^{-5} . Small difference between inclusions along diagonal centerline.

Fig. 9: Reconstruction using the Sec. 2.3 methodology (spatially varying regularization). Target from Fig. 1(a). Shear modulus contrast improved along diagonal centerline.

modulus values, when the inclusions are positioned diagonally as given in Fig. 1(c). On the other hand, the shape of the inclusions is well recovered as observed before. The compression on the top edge varies linearly as described in Sec. 2, thus the strain in the left inclusion will be much smaller than in the right inclusion. The shear modulus reconstructions of the left and right images in Fig. 5 are computed for a regularization factor of 4.0×10^{-10} and 1.0×10^{-10} . The left inclusion is clearly underestimated as can be seen in the shear modulus curves plotted over the centerline passing through the center of both inclusions.

3.2 Results for the Iterative, Regularized Inverse Problem Utilizing a Spatially Weighted Displacement Correlation.

The modified inverse problem statement, weighing the displacement correlation term with the reciprocal strain distribution, has been introduced in Sec. 2.2. We will show that this formulation reduces the sensitivity of the reconstructions presented earlier.

Figure 6 shows the shear modulus reconstruction obtained for the target shear modulus distribution given in Fig. 1(a) and the measured displacement data from the linear compression boundary. The shape of the inclusion is well recovered and the difference in the shear modulus values between both inclusions reduced significantly. The regularization factor has been chosen to be 3×10^{-5} .

Figure 7 represents the shear modulus reconstruction for the target shear modulus distribution defined in Fig. 1(b) with two inclusions having distinct stiffness values. A regularization factor of 7.0×10^{-6} is chosen, leading to a smooth shear modulus distribution. The inclusions are well recovered, and the relative contrast loss in each inclusion is more consistent as further discussed later.

The novel inverse method also works well for the diagonally positioned inclusion model with linear boundary compression (see Fig. 8). The difference in the shear modulus values between both inclusions is very small as can be clearly seen in the curve plot of the shear modulus values along the diagonal line passing through the center of both inclusions. A regularization factor of 2.5×10^{-5} has been chosen in the inverse problem solution.

3.3 Results for the Iterative, Regularized Inverse Problem Utilizing a Spatially Varying Regularization Factor.

The regularization factor of the previous results in Sec. 3.2 was intentionally chosen to be large. While a large regularization factor leads to very smooth and well recovered inclusion shapes, it compromises the overall contrast of the shear modulus values. In this section, we will utilize the results from Sec. 3.2 and perform a

[Figures 10, 11 — placeholders]

Fig. 10: (Left) Shear modulus reconstruction utilizing the Sec. 2.3 methodology. Target from Fig. 1(b). (Right) Centerline plot — contrast in both inclusions improved significantly, background smooth.

Fig. 11: (Left) Shear modulus reconstruction utilizing Sec. 2.3 methodology. Target from Fig. 1(c). (Right) Diagonal centerline plot — contrast improved without compromising shape quality.

posterior inverse solution step to improve the contrast, while retaining the smoothness quality of the reconstructions. For this, we update the regularization factor such that it becomes a spatial function defined in Sec. 2.3.

Applying this methodology to the target shear modulus distribution given in Fig. 1(a) with linear compression boundary results in a significant increase in the shear modulus contrast in the inclusions as shown in Fig. 9. We observe that the background is very smooth overall. We have selected a regularization factor of 3.5×10^{-5} for this case.

The shear modulus reconstruction corresponding to the target distribution in Fig. 1(b) is given in Fig. 10. We observe that the shear modulus contrast in both inclusions improves significantly, while the background remains smooth and the inclusion shapes are well recovered.

Finally, the shear modulus reconstruction for the target shear modulus distribution in Fig. 1(c) with diagonal positioned inclusions is given in Fig. 11, utilizing the inverse solution introduced in Sec. 2.3. Again, the shear modulus contrast improves significantly without compromising the quality of the recovered inclusion shape and the overall smoothness of the shear modulus distribution.

4 Discussion

In this paper, we have presented a novel formulation to solve the regularized inverse problem, posed as a constrained minimization problem subject to the constraint of the equilibrium equations. In a first step, we have modified the displacement correlation term, introducing a weighting function in terms of the strain components. We have tested this new methodology successfully with hypothetical data determined from three target shear modulus distributions given in Fig. 1. All these examples have in common that they comprise two inclusions in a homogeneous background. In the first example (see Fig. 1(a)) with two horizontally positioned inclusions of same stiffness value, we have shown that the conventional inverse method recovers the shear modulus distribution well (see Fig. 2) when a uniform displacement compression is applied. However, changing the boundary to a linear displacement compression, e.g., collecting ultrasound data while the ultrasound transducer is tilted, the recovered shear modulus values in both inclusions are significantly different. One could now conclude that the linear displacement compression results in small displacements in the left inclusion and large displacements in the right inclusion, thus are treated unequally in the displacement correlation term [45]. However, in the second example, the inclusions are horizontally positioned and have distinct stiffness values, while a uniform displacement compression is applied on the top boundary and the inverse problem is solved utilizing the conventional approach. We note that in this example, the displacements in both inclusions are similar, while the strain is smaller in the stiffer right inclusion. We expect that the reconstructed shear modulus values in both inclusions are underestimated consistently. In other words, the relative difference of target and reconstructed

shear modulus in each inclusion should be about same. This requirement is clearly not satisfied in the shear modulus reconstructions in Fig. 4 determined with the conventional inverse method. We note that it is not correct to say that the left inclusion is “better” recovered than the right inclusion. This is because in practical applications, we do not know the actual target modulus distribution, thus the reconstructions can only be properly interpreted if they are consistently underestimated. The novel proposed formulation addresses this issue successfully as shown in Fig. 7. In the last example (see Fig. 1(c)), we position the target inclusions with same stiffness values diagonally and apply a linear displacement compression on the top boundary. We note that this causes strains similar to the example in Fig. 1(a), but the displacements in the inclusions are significantly different. Again, the conventional method yields significant differences in the reconstructed shear modulus values in both inclusions. In particular, the inclusion with the lower strain has the lower reconstructed shear modulus value (see Fig. 5). This difference in the reconstructed shear modulus is significantly reduced utilizing the novel inverse formulation (see Fig. 8).

The examples utilized in this paper clearly demonstrate that the strain field is a potential candidate that is likely to influence the reconstruction of μ . To gain confidence in our shear modulus reconstructions and further investigate key variables in the new formulation, we derive a 1D analog given in Fig. 12. Two nonhomogeneous bars in parallel are connected by two rigid plates,

[Figure 12 — placeholder]

One-dimensional analogue diagram: two vertical nonhomogeneous bars side by side, connected by rigid plates (top and bottom, hatched). Each bar has a black inclusion region of length a^i ($i = 1, 2$) and background regions of length b^i above and below. Top plate applies displacements u^1 and u^2 to the left and right bars respectively. Total bar length = 1 (unit). Labels: a^1, b^1 on left bar; a^2, b^2 on right bar.

Fig. 12 One-dimensional analogue of previous shear modulus inclusions, represented by two nonhomogeneous bars



where the bottom plate is fixed and the top plate can be tilted to apply different compression levels on each bar, denoted by u^i . The length a^i ($i = 1, 2$) is analog to the stiff inclusion diameter. The stiffness value in the inclusions will be chosen according to the cases discussed earlier and the remaining region of the bar will have a stiffness value of 1. The objective function to be minimized is

$$F = \sum_{i=1}^2 F^i = \sum_{i=1}^2 \left(\frac{1}{2} \int_0^1 (u_{\text{com}}^i - n \cdot u_{\text{exact}}^i)^2 dy + \alpha |\mu_{in}^i - \mu_b^i| \right) \quad (4.1)$$

where F^1 and F^2 are contributions to the objective function from the left ($i = 1$) and right ($i = 2$) bars, respectively. In Eq. (4.1), u_{com}^i and u_{exact}^i denote the computed and exact displacement fields, respectively, and n is a factor to test our analysis for sensitivity (similar to noise). The second term, TVD regularization, is analytically expressed in 1D for the nonhomogeneous bar, where μ_{in}^i is the inclusion stiffness and μ_b^i is the background stiffness. We may set $\mu_b^i = 1$ as the reconstructions are purely relative, i.e., off by a multiplicative factor. Next, we express the displacement field given that the strain is piecewise constant in the background and inclusion

$$u_{\text{com}}^i = \begin{cases} b^i \cdot \varepsilon_b^i + a^i \cdot \varepsilon_{in}^i + (y - b^i - a^i) \varepsilon_b^i & \text{when } b^i + a^i < y \leq 1 \\ b^i \cdot \varepsilon_b^i + (y - b^i) \varepsilon_{in}^i & \text{when } b^i < y \leq b^i + a^i \\ y \cdot \varepsilon_b^i & \text{when } 0 < y \leq b^i \end{cases} \quad (4.2)$$

where ε_b^i and ε_{in}^i are the computed strains in the background and the inclusion, respectively, and the indices $i = 1, 2$ denote the left and right inclusion. Similarly, the exact displacement field u_{exact}^i can be expressed by replacing the strains in Eq. (4.2) by the exact strains $\bar{\varepsilon}_b^i$ and $\bar{\varepsilon}_{in}^i$. As the stress in each bar is constant, we have the relationships $\varepsilon_b^i = \mu_{in}^i \cdot \varepsilon_{in}^i$ and $\bar{\varepsilon}_b^i = \bar{\mu}_{in}^i \cdot \bar{\varepsilon}_{in}^i$ where the bars denote the exact variables. From this relationship and kinematics, we obtain

$$\varepsilon_b^i = u^i / ((1 - a^i) + a^i / \mu_{in}^i) \quad \text{and} \quad \bar{\varepsilon}_b^i = \bar{u}^i / ((1 - a^i) + a^i / \bar{\mu}_{in}^i) \quad (4.3)$$

where u^i and $\bar{u}^i$ are the noisy and exact displacement loadings applied on the top of the bars. Using Eqs. (4.2)–(4.4), the objective function in Eq. (4.1) can be expressed in terms of the unknown μ_{in}^i ; thus, the gradient can be directly determined by

$$\frac{\partial F}{\partial \mu_{in}^i} = \frac{\partial F^1}{\partial \mu_{in}^i} + \frac{\partial F^2}{\partial \mu_{in}^i} = 0 \quad (4.4)$$

It is important to acknowledge that F^1 does not depend on μ_{in}^2 and F^2 does not depend on μ_{in}^1 . Thus, Eq. (4.4) can be decoupled and minimized separately for each bar

$$\frac{\partial F^i}{\partial \mu_{in}^i} = \frac{f^i}{3 (\mu_{in}^i (-1 + a^i) - a^i)^3 (\bar{\mu}_{in}^i (-1 + a^i) - a^i)} + \alpha = 0, \quad i = 1, 2 \quad (4.5)$$

where f^i is a lengthy expression given in Eq. (4.6) involving u^i , a^i , b^i , μ_{in}^i , $\bar{\mu}_{in}^i$, and the noise factor n .

[Equation (4.6) — a six-term expansion of f^i involving products of u^i , $(a^i)^2$, μ_{in}^i , $\bar{\mu}_{in}^i$, b^i , and polynomial expressions in a^i and b^i . Transcription omitted due to length and pdftotext rendering artifacts; see PDF page 7 for the exact form.]

As the regularization factor in the left and right bars is the same, we obtain from Eq. (4.5)

$$\frac{f^1}{3 (\mu_{in}^1 (-1 + a^1) - a^1)^3 (\bar{\mu}_{in}^1 (-1 + a^1) - a^1)} = \frac{f^2}{3 (\mu_{in}^2 (-1 + a^2) - a^2)^3 (\bar{\mu}_{in}^2 (-1 + a^2) - a^2)} \quad (4.7)$$

Similar to Eq. (4.7), one can obtain the expression for the spatially weighted displacement correlation term (not shown here). Next, we fix the right stiffness value to the reconstructed value observed in the 2D model for the cases discussed in Sec. 3 and solve for the stiffness modulus in the left inclusion. The results in Tables 1 and 2 clearly show that our 1D analog, while simple, can closely reproduce the trends observed in the 2D inverse solution, thus confirms that the observations in the 2D model are not due to convergence related issues. Furthermore, we note that the observed trends are independent of perturbations (noise) applied to the displacement field. We have made this same observation in our 2D models (not shown in the manuscript). Thus, we conclude that the regularization term drives these ambiguous reconstructions and not the displacement noise level.

Table 1

Comparison of reconstructions for left inclusion with 2D model, 1D model, and 1D model with noise. The conventional objective function is used.

	2D continuum model		1D bar model ($n = 1$)		1D bar model ($ n - 1 = 1.5\%$)	
	Left incl.	Right incl.	Left incl.	Right incl.	Left incl.	Right incl.
Case 1	2.2	2.98	2.06	2.98	2.07	2.98
Case 2	4.86	6.44	4.39	6.44	4.37	6.44
Case 3	2.15	3.01	2.08	3.01	2.21	3.01

Table 2

Comparison of reconstructions for left inclusion with 2D model, 1D model, and 1D model with noise. The spatially weighted objective function is used.

	2D continuum model		1D bar model ($n = 1$)		1D bar model ($ n - 1 = 1.5\%$)	
	Left incl.	Right incl.	Left incl.	Right incl.	Left incl.	Right incl.
Case 1	2.69	2.95	2.95	2.95	2.96	2.95
Case 2	4.11	6.34	3.66	6.44	3.64	6.44
Case 3	2.69	2.95	2.95	2.95	3.06	2.95

To motivate the spatial weighted factor as a function of strain, we discard noise as it has limited influence on the observed results. This significantly simplifies the following analysis. Furthermore, we assume that the left and right inclusions have the same diameter of 0.2 and the inclusions can be positioned such that the relationship $b_1 = 0.8 - b_2$ holds, which will cancel out terms dependent on b_1 and b_2 . Thus, Eq. (4.7) simplifies to

$$\frac{(\mu_{in}^1 - \bar{\mu}_{in}^1)(\mu_{in}^2(-1+a) - a)^3}{(\mu_{in}^2 - \bar{\mu}_{in}^2)(\mu_{in}^1(-1+a) - a)^3} = \frac{(\bar{u}^2)^2(\bar{\mu}_{in}^1(-1+a) - a)}{(\bar{u}^1)^2(\bar{\mu}_{in}^2(-1+a) - a)} = \frac{\bar{u}^2 \varepsilon_{in}^2}{\bar{u}^1 \varepsilon_{in}^1} \quad (4.8)$$

where a is the left and right inclusion size with $a = 0.2$. The right-hand side of Eq. (4.8) follows from kinematics and equilibrium. We fix $\mu_{in}^2 = 0.8\bar{\mu}_{in}^2$ assuming a 20% contrast loss in the right inclusion, and solve for the shear modulus variable μ_{in}^1 in the left inclusion. It is important to acknowledge that $\bar{u}^1$ can be applied independently of $\bar{u}^2$, thus from Eq. (4.8) we realize that solving for μ_{in}^1 will strongly depend on this displacement boundary. The expression for the spatially weighted regularization case is given by

$$\frac{(\mu_{in}^1 - \bar{\mu}_{in}^1)(\mu_{in}^2(-1+a) - a)^3}{(\mu_{in}^2 - \bar{\mu}_{in}^2)(\mu_{in}^1(-1+a) - a)^3} = \frac{(1 + ((\bar{\mu}_{in}^1)^2 - 1)a)(\bar{\mu}_{in}^1(-1+a) - a)}{(1 + ((\bar{\mu}_{in}^2)^2 - 1)a)(\bar{\mu}_{in}^2(-1+a) - a)} \quad (4.9)$$

Equation (4.9) clearly shows that the solution for μ_{in}^1 does not depend on the applied boundary condition using the spatially weighted formulation. As the contrast in the right inclusion has been assumed to be underestimated by 20%, we expect the left inclusion to be underestimated by 20% as well. Keeping this in mind, we plot the error defined by $|(\mu_{in}^1 - 0.8\bar{\mu}_{in}^1)/(0.8\bar{\mu}_{in}^1)| \times 100\%$ over $\bar{\mu}_{in}^2$ for different $\bar{\mu}_{in}^1$ ($= 3, 4, 5, 6, 7, 8, 9, 10$ from top curve to bottom curve) for the conventional method in Fig. 13 and the spatially weighted displacement correlation term in Fig. 14. The displacement boundary ratio between right and left boundaries is set to 2. We observe that the error for the conventional method reaches about 70%, while the reconstruction error for the new method is about 22%. The plots also reveal that the error for the new method is zero when the stiffness values in left and right inclusions are same, while for the conventional method it is about 25%. We note that the error plot will change for different displacement boundaries using the conventional method, but will remain the same for the spatially weighted displacement correlation term. We also analyzed the case with uniform compression for the conventional method and the spatially weighted formulation in Fig. 15. We observe that the maximum error is about 55% in the shear modulus reconstruction. Overall, the spatially weighted formulation appears to be superior to the conventional method for this analysis. Finally, this approach can be generalized for more than two bars, i.e., multiple bars with inclusions in parallel, and the above analysis will hold as each bar can be decoupled in the minimization analysis.

[Figures 13, 14 — placeholders]

Fig. 13: Error plot, conventional method. Curves for $\bar{\mu}_{in}^1 = 3, 4, \dots, 10$ (top to bottom). X-axis: shear modulus $\bar{\mu}_{in}^2$ (range 3–10). Y-axis: Error in μ (%). Errors reach $\sim -60\%$ to -70% for large $\bar{\mu}_{in}^2$. All curves negative (underestimation).

Fig. 14: Error plot, spatially weighted method. Same curve family. Errors capped at $\sim -22\%$, zero when $\bar{\mu}_{in}^1 = \bar{\mu}_{in}^2$ (equal stiffness). Significant improvement.

[Figure 15 — placeholder]

Error plot of the reconstructed shear modulus in the left inclusion over the exact shear modulus in the right inclusion $\bar{\mu}_{in}^2$ for different shear modulus values in the left inclusion $\bar{\mu}_{in}^1$ ($= 3, 4, 5, 6, 7, 8, 9, 10$). The conventional method has been used for the case that the compression is applied uniformly. Curves all negative, reaching $\sim -55\%$ at the extremes.

the novel inverse formulation in Sec. 2.2. In Sec. 2.3, we have introduced a posterior inverse formulation that utilizes the smooth shear modulus distribution obtained in Sec. 3.2 to compute an updated spatially varying regularization factor. To motivate the spatially varying regularization factor, we first acknowledge that for $\alpha = 0$ (no regularization term), the shear modulus reconstruction would strongly oscillate. The oscillations will have smaller amplitudes in the stiffer inclusion than in the background. This can also be observed from the reconstructions presented with the smaller choice of the regularization factor for the conventional approach. Thus, we expect that the shear modulus will be penalized more in the inclusion than in the background. Spatially regularizing the problem will reduce this effect. Figures 9–11 reveal that this posterior inverse solution strategy increases the shear modulus values in the inclusion by about 30% without compromising the overall smoothness of the reconstructions. In order to compare the shear modulus reconstruction from the conventional inverse approach, the regularization factor was selected such that the overall shear modulus contrast is similar and comparable to the reconstructions of the spatially varying regularization factor. Clearly, the posterior inverse approach is superior to the conventional inverse methodology, which has strong oscillations in the background.

5 Conclusions

We have shown that the conventional regularized inverse problem formulation posed as a minimization problem results in ambiguous shear modulus reconstructions and depends on the choice of boundary conditions. We have observed this on simulated data created by solving the forward problem for a predefined target shear modulus distribution. We have reduced the boundary sensitivity of the regularized inverse solution by weighting the displacement correlation term with the inverse strain tensor having only normal strain components. We have derived a 1D analog and shown analytically that our inverse strategy will work for an arbitrary number of same sized inclusions with different stiffness values. The 1D model analysis has the limitation that it does not account for inclusions sharing the same horizontal coordinate (i.e., vertically shifted), though it can be expanded to include these cases as well. Although our reconstructions clearly show the limitations of the conventional approach, the spatial weighted method needs to be investigated further. We also note that these observations were analyzed for total variation regularization and may not hold for other types of regularizations. However, the results and analysis in this paper provide a better understanding in the solution of the regularized inverse problem, and may help to analyze other regularization types in a similar fashion. Finally, we have drastically improved the contrast of the shear modulus reconstructions without compromising the overall quality. This has been achieved in a posterior inverse solution step, expressing the regularization factor as a spatial function of the coordinates rather than as a constant.

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